FLORENCE

Multimodal AI Gym Assistant

Project Documentation & Manual Test Report

Project: Call It A Day: Gym & Lifestyle Club
Platform: n8n + Telegram + OpenAI + Supabase Vector Store + Google Drive
Prepared for: Mohamed Saad Master
Evidence window: 25–27 October 2026
Report version: 1.0 — compiled from supplied workflow exports, dataset, screenshot and test transcript

This report documents the supplied configuration and observed test behavior. It distinguishes facts visible in the supplied artifacts from interpretation and known gaps.

1. Executive summary

Florence is a Telegram-based assistant for the fictional Call It A Day Gym & Lifestyle Club in Wicker Park, Chicago. It accepts text, voice notes and photos; converts each input into a common text field; passes the resulting context to an AI Agent; retrieves club information from a Supabase vector store; and sends a text response to Telegram. A separate n8n workflow watches a Google Drive folder and inserts newly created documents into the same vector-store table.

The project combines retrieval-augmented generation (RAG) with multimodal input handling. The business dataset is intended to be the source of truth for club-specific facts such as prices, opening hours, services and policies. Image analysis is used for visible content, while database retrieval is intended to verify whether a pictured facility or service actually belongs to the club.

> **Important status note:** both supplied workflow JSON exports contain `"active": false`. This means the exported configurations are marked inactive; it does not prove whether a different saved copy was active at another time.

What is included

• Project overview and architecture

• Node-by-node descriptions of both supplied n8n exports

• The supplied Telegram Agent canvas screenshot

• A clearly marked placeholder for the RAG Pipeline canvas screenshot

• Business-data reference summary and key facts

• Manual test matrix based on the supplied Telegram transcript

• Observed failures, refinements, limitations, security notes and recommended next steps

Evidence and scope

This report is based on the complete gym dataset pasted in the conversation, the Telegram Agent JSON export, the RAG Pipeline JSON export, one Telegram Agent canvas screenshot, and the supplied manual chat transcript dated 25–27 October 2026. It is not a formal benchmark or a live inspection of the n8n instance. The complete raw JSON text is retained in the conversation; this PDF explains the exports and their relevant configuration rather than reproducing every credential and instance identifier.

2. Project goals and scope

Primary goal

Provide a friendly, concise Telegram assistant that answers questions about gym membership, services, prices, schedules, facilities and policies using the club's source data, while also responding to voice notes and images.

In scope

• Telegram message intake and routing for text, voice and photo messages.

• Voice transcription into a shared user_text field.

• Image analysis for food, gym equipment and facilities, with image analysis passed to the agent alongside the customer caption.

• AI Agent with a language model, short-window memory and a Supabase retrieval tool.

• Google Drive folder polling and ingestion of newly created documents into a Supabase vector store.

• Manual exploratory testing through Telegram.

Not demonstrated by the supplied exports

• Automated appointment booking, payment processing or reservation confirmation.

• A staff escalation integration that actually contacts the team. The fallback wording offers follow-up, but no team-contact node is shown.

• Update/delete synchronization for files already ingested into Supabase.

• Automated regression tests, load tests, monitoring dashboards or a formal scoring benchmark.

• Proof that the currently deployed workflows match these exact exported versions.

3. High-level architecture

The two workflows have separate responsibilities: the RAG Pipeline builds the searchable knowledge base; the Telegram Agent handles customer conversations and queries that knowledge base.

Layer	Components	Purpose
Knowledge ingestion	Google Drive Trigger → Download file → Default Data Loader → Recursive Character Text Splitter → OpenAI Embeddings → Supabase Vector Store	Convert newly created source files into embedded document chunks in the documents table.
Conversation intake	Telegram Trigger → Switch	Detect message type and route voice, photo and text inputs.
Input normalization	Edit Fields / Edit Fields1 / Edit Fields2	Normalize text, transcript or caption + image analysis into user_text.
Reasoning and retrieval	AI Agent + OpenAI Chat Model + Simple Memory + gym_database	Compose a response using the prompt, conversational context and retrieved club information.
Response	Send a text message	Send the agent output to the Telegram chat.

RAG in this project

Retrieval-augmented generation is the pattern of retrieving relevant source text and supplying it to a generative model at answer time. Here, the vector-store tool is named gym_database and queries the Supabase table documents. The Telegram Agent export sets topK to 6. The ingestion export configures OpenAI embeddings and inserts documents into the same table. This architecture is intended to ground club-specific answers in the supplied dataset; the prompt alone cannot guarantee correct retrieval or perfect adherence.

4. Workflow A — Telegram Agent

Trigger and routing

Telegram Trigger listens for message updates. The Switch has three named outputs: voice, photo and text, with an additional fallback output named extra. The supplied connection map routes the first three outputs, but no connection is shown for extra. As a result, unsupported message types may end without a user-facing response.

Text branch

The text branch uses Edit Fields to set user_text to the Telegram message text, then passes the item to AI Agent.

Voice branch

The voice branch downloads the Telegram voice file, sends the recording to the OpenAI transcription node, then uses Edit Fields1 to prefix the transcript with “Customer sent a voice note: ” and store it as user_text. The common AI Agent then handles the resulting text.

Photo branch

The photo branch requests the last/highest-resolution entry in Telegram’s message.photo array, downloads that file, and passes it to the OpenAI image-analysis node using GPT-4o-mini. Its instructions ask it to identify visible foods and estimate calories/protein, identify gym equipment and safe use, or describe fitness-related imagery while avoiding diagnoses and unsupported business claims. Edit Fields2 combines the Telegram caption (or “No caption provided”) with the image-analysis text. That combined user_text is sent to AI Agent.

Agent and response

AI Agent uses the OpenAI Chat Model configured as gpt-5-mini, Simple Memory keyed by Telegram chat ID with contextWindowLength 4, and the Supabase Vector Store named gym_database in retrieve-as-tool mode with topK 6. The final Send a text message node sends $json.output to the original Telegram chat ID.

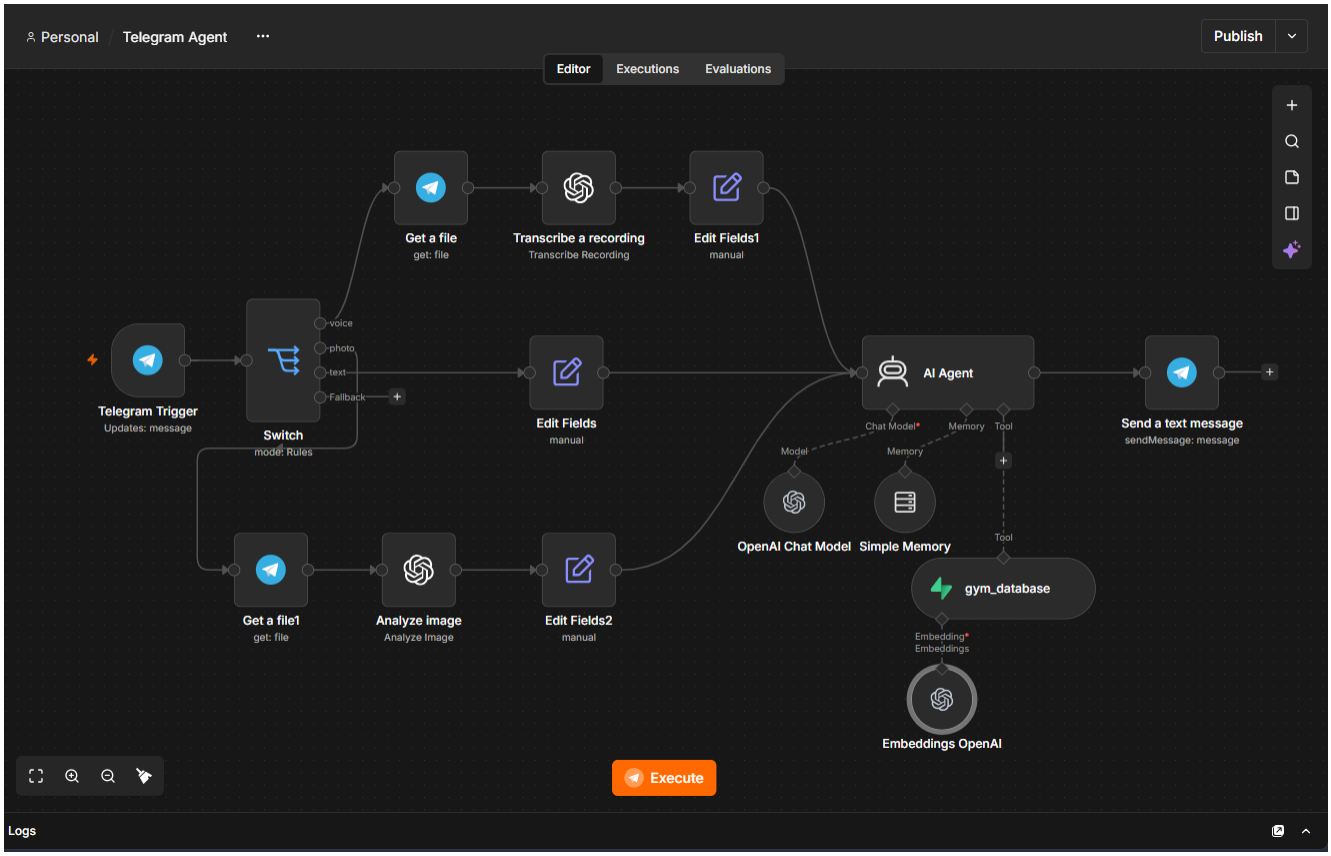
Prompt design

The system prompt establishes Florence’s role, requires retrieval for business-specific facts, specifies a literal fallback for unknown information, asks for concise and relevant answers, cautions against unsupported future announcements, and gives special instructions for photos, food estimates, gym equipment, health concerns, and bookings. It correctly says not to claim a booking is confirmed without a booking system. However, the transcript shows that prompt instructions did not eliminate every failure: the model still introduced a false “swimming lessons are coming soon” claim in one response and sometimes included unrelated or excessive details.

5. Telegram Agent canvas screenshot

Figure 1. Supplied n8n canvas screenshot for the Telegram Agent workflow.

The canvas visibly shows the three input branches feeding the AI Agent, the connected model, memory and gym_database tool, and the Telegram reply node. Node names and wiring are interpreted together with the supplied JSON export.



6. Workflow B — RAG Pipeline

Trigger and file download

Google Drive Trigger watches the specific folder named RAG DATA, polls every minute, and is configured for the fileCreated event. Download file uses the trigger item's id as the fileId.

Document preparation

Default Data Loader is configured for binary data and custom text splitting. It attaches a metadata field named file_name using the name from Google Drive Trigger. Recursive Character Text Splitter connects to the loader through the ai_textSplitter connection and has chunkOverlap set to 200. The chunk size is not explicitly specified in the supplied JSON and should not be guessed.

Embedding and storage

Embeddings OpenAI connects to Supabase Vector Store through the ai_embedding connection. Supabase Vector Store is in insert mode and targets the documents table. The embedding node's options are empty, so the exact embedding model is not identified in this export.

Known scope limitation

The trigger listens for file creation only. No update, deletion, duplicate-checking, re-indexing or cleanup path appears in this export. A file edited after ingestion may not refresh the stored chunks automatically, and repeated ingestion behavior should be tested before relying on this as a maintained knowledge base. The export is marked inactive.

7. RAG Pipeline canvas screenshot

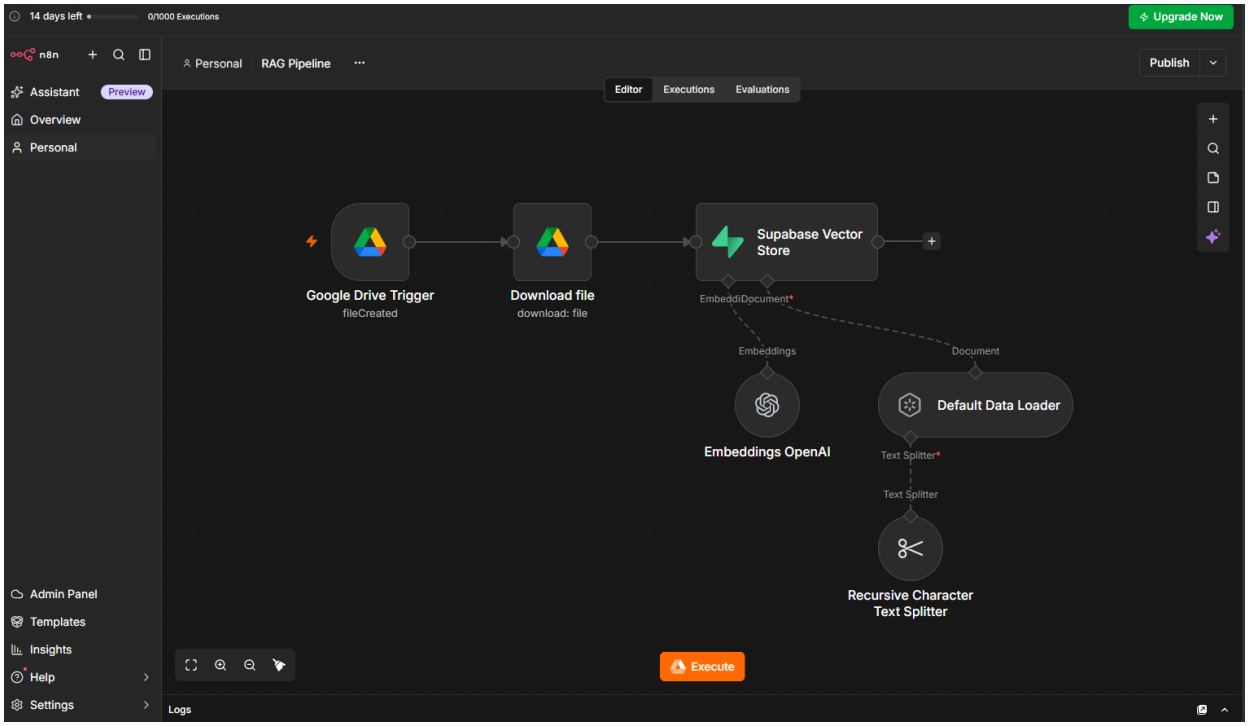


Figure 2. Insert the RAG Pipeline canvas screenshot here.

Placement note: replace this placeholder with a screenshot of the RAG Pipeline canvas. For legibility, crop to the workflow editor and include the node names and connection lines. Avoid exposing API keys, tokens, account IDs or private webhook values.

8. Source dataset — business facts

The supplied Call It A Day Gym Data File is a 10-page fictional dataset. Its address, phone number and prices are dummy data for this project and should not be treated as real-world business information. The full source text was supplied in the conversation. The following reference summarizes the key facts most relevant to testing; the source dataset remains authoritative for details not repeated here.

Club overview

Call It A Day is described as a full lifestyle gym in Wicker Park, Chicago, with gym floor, Pilates, sauna, massage chairs, smoothie bar, sports physiotherapy, nutrition coaching, barber shop and supervised Kids Zone. Address: Suite 200, 1425 N Milwaukee Ave, Chicago, IL 60622. Phone/text: (312) 555-0142. Email: hello@calliteaday.example. Instagram: @calliteadayfit. Nearest transit: 2-minute walk from Damen Blue Line. Members receive two hours of validated parking in the neighboring garage; bike racks are inside the lobby.

Opening hours

Area	Hours
Gym — Mon–Fri	5:00 AM–11:00 PM
Gym — Saturday	6:00 AM–9:00 PM
Gym — Sunday	7:00 AM–8:00 PM
Kids Zone — Mon–Fri	8:00 AM–12:00 PM; 4:00 PM–8:30 PM
Kids Zone — weekends	9:00 AM–5:00 PM
Sauna & Smoothie Bar — Mon–Fri	6:00 AM–10:00 PM
Sauna & Smoothie Bar — Saturday	7:00 AM–8:00 PM
Sauna & Smoothie Bar — Sunday	8:00 AM–7:00 PM
Holidays	Closed Thanksgiving and Christmas; 8:00 AM–4:00 PM New Year's Day and July 4

Membership prices (selected reference values)

Plan	1 month	3 months	12 months	Included
Basic	$59	$159	$499	Gym floor
Plus	$89	$239	$749	Gym floor, sauna, massage chairs
All Access	$139	$379	$1,199	Plus benefits, unlimited Pilates, Kids Zone (up to 3 h/visit), 4 smoothies/month
Student	—	$129	—	Gym floor; valid student ID; off-peak weekdays 5 AM–4 PM and all day weekends

Additional membership facts: $49 one-time joining fee unless waived during offers; gym-only day pass $20; gym + sauna day pass $30; first-time visitors can book one free 60-minute gym-floor trial in advance. Couple/family annual plans are $1,099 for two people on Plus or $2,099 for two people on All Access.

9. Dataset service and policy reference

Personal training

Certified one-on-one training: $75 single session; 12-pack $840 ($70/session); 36-pack $2,160 ($60/session). Every new member receives a free fitness assessment with a trainer. Training packs are non-refundable and usable within six months.

Pilates

Mat Pilates: Monday, Wednesday and Friday at 7:00 AM and 6:30 PM; Saturday at 9:00 AM. Reformer Pilates: Tuesday and Thursday at 7:30 AM and 7:00 PM; Saturday at 10:30 AM. Class capacity is eight. All Access includes unlimited classes. Other members pay $25 per mat class or $35 per reformer class; 10-class packs cost $220 (mat) or $320 (reformer). Booking required; arrive 10 minutes early; cancellation allowed up to four hours before class.

Sauna and massage chairs

Two private dry-heat sauna rooms, each fitting up to four people, with 20-minute sessions. Included with Plus and All Access; Basic members can use the $30 gym-plus-sauna day pass. Shower before entry, bring a towel and avoid use immediately after a heavy meal. People with heart conditions, high blood pressure or pregnancy should check with a doctor first. Four full-body massage chairs offer 15-minute sessions; included with Plus and All Access, or $8 per session for Basic members.

Smoothie bar

Berry Blast $9; Mango Lassi Protein shake $10; Peanut Butter Banana shake with whey $11; Green Detox $10; Chocolate Whey $10; Cold brew protein shake $11. Extra protein scoop $2.50. Oat or almond milk vegan options cost no extra. All Access includes four smoothies per month up to $11 each. Open to members and non-members.

Kids Zone

Ages 3–10, two trained staff, maximum 15 children, maximum three hours per visit. Parent/guardian must remain on premises. Free for All Access and Family members; otherwise $10/hour/child. Registration requires child name, age and parent phone number; allergies should be disclosed. Peanut-free; meals not included; children under three cannot be accepted.

Sports physiotherapy and nutrition coaching

Physiotherapy: initial 60-minute assessment $120; 45-minute follow-up $90; five follow-ups $400. Members receive 15% off; annual All Access members get one free mobility screening. Hours Mon/Wed/Fri 8 AM–6 PM, Sat 9 AM–2 PM; closed Sun/Tue/Thu. No referral needed; no direct insurance billing, but receipts are available. Nutrition: first 60-minute consultation $75; 30-minute follow-up $40; four-week package $160 including consultation, personalized plan and two follow-ups. Annual All Access includes one free first consultation. Hours Tue/Thu 10 AM–7 PM and Sat 10 AM–2 PM; video sessions available.

10. Dataset service and policy reference (continued)

Barber shop and events

Barber prices: haircut $40; fade/skin fade $45; beard trim $20; haircut + beard trim $55; hot towel shave $35; children under 12 $25. Members receive 10% off. Hours Tuesday–Saturday 10 AM–7 PM, Sunday 10 AM–4 PM, closed Monday; walk-ins welcome. Events include a monthly fitness challenge (first Saturday), weekend workshops usually $20–$40 and free for All Access, a monthly Sunday Social, and quarterly Kids Fun Day. Dates are announced on Instagram and the member text group.

Merchandise

The source labels branded merchandise “Coming Soon” and says launch date and prices are not announced. This specific source statement is valid dataset content, but the agent must not extrapolate that label into announcements about unrelated services such as swimming lessons.

Policies

Memberships are non-refundable once started. Within 72 hours of purchase, if the gym has not been used, a refund is available minus the joining fee. Three-month plans can freeze up to seven days; annual plans up to 30 days in total in blocks of at least seven days. Requests must be made at least two days ahead. Medical freezes with a doctor’s note may extend to 60 days for annual plans. Membership can be transferred once for $50; upgrades are allowed by paying the difference for the remaining period; downgrades are not allowed during an active membership. Cards, Apple Pay, Google Pay, ACH bank transfer and cash are accepted. Annual plans can be paid in 12 equal monthly card payments at no extra cost. Taxes apply where required.

Rules

Towel required for sauna; clean indoor sports shoes; wipe equipment after use; minimum gym-floor age 16 (parental consent for under 18); no outside food on the gym floor, water bottles allowed; lockers are day-use only.

Unknown-information fallback

The agent prompt specifies: “I don't have that detail right now, but our team will confirm and get back to you.” No tool or integration in the supplied workflow actually contacts the team, so this wording should be understood as a conversational fallback rather than a confirmed escalation action.

11. Manual test report — 25 October 2026

The transcript begins with /start and greeting tests, then tests personal-training and nutrition pricing, opening hours and unsupported swimming lessons. Ratings below reflect the observed transcript, not an independent rerun.

ID	Input / scenario	Observed result	Assessment
T01	/start / “Hi” / “Hello”	Brief greeting and asks how it can help.	Pass
T02	Personal trainers for fat loss and dietitian	Correct training and nutrition prices; mentions new-member assessment and annual All Access nutrition consultation. Some response verbosity.	Pass with note
T03	Cost of 12 PT sessions and new-member inclusion	Correct $840 / $70 per session and free fitness assessment. Adds joining fee/membership information not explicitly requested.	Pass with relevance issue
T04	Voice note asking PT for fat loss and cost	Correct prices and free assessment returned.	Pass
T05	Opening hours	Correct weekly gym hours and holiday exceptions.	Pass
T06	Swimming lessons	First reply states the service is not listed and uses the required fallback. Later response claims lessons are “coming soon” without source support.	Fail; hallucination observed

12. Manual test report — 27 October 2026

ID	Input / scenario	Observed result	Assessment
T07	Voice message asking for protein guidance	General protein ranges and medical caveat.	Partial: nutrition answer delivered, but not a club-specific query
T08	Food photo: grilled chicken, sweet potato, cucumber, cilantro	One response estimated about 290–410 kcal and 27–34 g protein with portion assumptions. An earlier reply failed to answer the meal-photo request.	Partial; retry improved
T09	Food photo asks itemized calories/protein per item and total	Reply gave a general estimate for a different-looking plate (salmon, avocado, rice, corn) but did not itemize each item or state portion assumptions.	Fail
T10	Leg-press photo: name, muscles, safe form, beginner mistakes	First reply focused on promotions and did not answer the whole question; repeat reply identified leg press, target muscles, technique and common mistakes.	Partial; retry improved
T11	Voice question about basketball court availability	Says court is not listed and uses fallback.	Pass
T12	Sauna photo and cost/access question	Later responses correctly describe sauna, access via Plus/All Access and $30 day pass for Basic.	Pass with relevance/verbosity issues
T13	Repeated sauna questions after swimming-lesson context	Swimming lessons repeatedly reappear, including unrelated follow-up prompts.	Fail: context relevance
T14	Sauna response includes unsupported “coming soon” swimming lessons	Contradicts dataset/prompt guardrail.	Fail
T15	Sauna response lists many membership durations and unrelated facts	Later versions become more concise, but some responses over-list plans and extra details.	Partial; iterative refinement

Interpretation: The transcript demonstrates that the core text/voice/photo paths produced replies and that factual answers could be accurate. It also documents meaningful failure modes in adherence, completeness, hallucination control and relevance. These are qualitative manual observations; no automated score, pass-rate claim or reproducible test harness was supplied.

13. Findings and recommended improvements

Priority 1 — prevent unsupported claims

The false “swimming lessons are coming soon” statement is the clearest factuality failure. Keep the explicit “never invent future plans” rule, but also consider requiring a retrieved source snippet before making any availability or launch claim. Add a negative test where swimming lessons are asked about repeatedly after unrelated topics. Verify the exact fallback is returned without adding speculative statements.

Priority 2 — isolate follow-up context

The memory window is configured to four messages, but the transcript shows unrelated swimming-lesson context recurring during sauna answers. Test a clean conversation, a conversation with unrelated earlier questions, and repeated facility questions. Consider reducing memory scope, summarizing only relevant facts, or explicitly instructing the agent not to answer stale topics unless asked. The transcript does not prove that memory adjustment fully fixed the issue.

Priority 3 — answer multi-part image questions completely

For food photos, enforce a structured answer: each visible item, assumed portion, calories, protein, and total. If the photo does not support reliable portion estimates, label uncertainty rather than praise a generic “balanced plate.” For equipment photos, answer name, target muscles, setup, technique, and beginner mistakes before any optional booking suggestion. Re-test the exact multi-part prompts.

Priority 4 — concise, relevant business answers

For facility pricing, lead with availability, applicable access price and only the membership prices relevant to access. Avoid listing all durations or unrelated benefits unless requested. Keep the system prompt’s “one relevant next step” principle consistent across duplicated prompt sections.

Priority 5 — RAG maintenance and routing

Add a user-facing fallback for the Switch extra output; define update/delete/duplicate handling for ingested files; validate file types and empty documents; and test retrieval for prices, dates, membership comparisons and policy edge cases. Consider including source metadata in retrieval outputs to improve traceability.

14. Security, privacy and operational considerations

Credential handling

The JSON includes credential reference IDs, workflow IDs, a Telegram webhook ID and a Google Drive folder identifier. These values are not passwords by themselves, but operational identifiers should not be published unnecessarily. Never include bot tokens, API keys, OAuth refresh tokens, private webhook URLs or Supabase service-role keys in the report or screenshots. The supplied report intentionally omits credential identifiers.

Data handling

Voice notes are sent to a transcription service, photos to an image-analysis model, and user text plus retrieved business content to the AI Agent/model. Review the privacy terms and retention settings of the connected services before using real customer data. The dataset is fictional; testing with real names, contact details, children's information, injury details or nutrition/health information would create additional privacy obligations.

Health boundaries

The prompt appropriately prohibits diagnoses and advises users with relevant medical conditions to consult qualified professionals. Food macro estimates should remain approximate and based on visible portions. Exercise technique advice should be general and should not be framed as personalized medical treatment.

Reliability and operations

The exports are marked inactive and rely on external credentials and service configuration. Before activation, test each branch, confirm the Telegram webhook/trigger setup, validate Supabase table/schema and embedding compatibility, and confirm expected model availability and billing. Add error handling for transcription, image analysis, retrieval and Telegram send failures.

15. Reproducibility checklist

Use this checklist when setting the project up or demonstrating it again. It describes checks to perform, not claims that they have already been completed.

- ■ Import both workflow JSON exports into the intended n8n instance.
- ■ Reconnect credentials securely: Telegram, Google Drive, OpenAI and Supabase.
- ■ Confirm the Google Drive RAG DATA folder and the files to ingest.
- ■ Verify the Supabase documents table and the expected vector extension/schema.
- ■ Check embedding model compatibility between ingestion and retrieval.
- ■ Run the RAG Pipeline manually with a small test document and verify chunks/metadata are inserted.
- ■ Test text, voice and photo branches independently through Telegram.
- ■ Test unsupported message types and confirm a fallback response.
- ■ Test business facts against the dataset, including prices, hours, holidays, discounts and policy edge cases.
- ■ Run negative tests for unsupported services, especially swimming lessons and other unlisted facilities.
- ■ Run image tests for food itemization and complete gym-equipment explanations.
- ■ Verify that unrelated conversation history does not leak into facility answers.
- ■ Confirm error paths, logging and user-facing error messages.
- ■ Only activate workflows after credentials, retrieval and all three input branches pass.
- ■ Re-run the regression suite after prompt, model, memory, splitter or dataset changes.

Suggested regression prompts

- “How much is the 12-session personal training pack, and what does a new member get?”
- “What are your Sunday gym and sauna hours?”
- “Do you offer swimming lessons?” (should not invent availability or future plans)
- “Identify each food item in this photo; estimate calories and protein per item and total, with assumptions.”
- “Name this machine, identify the main muscles, explain safe form and list beginner mistakes.”
- Ask about swimming lessons, then ask a new sauna pricing question; sauna answer should not resurrect swimming lessons.

16. Artifact inventory and evidence notes

Artifact	Status in this report	Notes
Call It A Day gym data file	Supplied in conversation; summarized in Sections 8–10	Full source dataset is 10 pages. Prices/address/contact are fictional placeholder values.
Telegram Agent JSON	Supplied in conversation; analyzed in Sections 4 and 15	Export name: Telegram Agent. active=false. Raw JSON remains in conversation.
RAG Pipeline JSON	Supplied in conversation; analyzed in Section 6	Export name: RAG Pipeline. active=false. Creation-only trigger.
Telegram Agent canvas screenshot	Included as Figure 1	Image supplied by user.
RAG Pipeline canvas screenshot	Placeholder included as Figure 2	Insert screenshot manually later.
Manual Telegram chat history	Analyzed in Sections 11–12	Transcript dated 25 and 27 October 2026.
Automated benchmark / execution logs	Not supplied	No numerical benchmark or independent execution evidence is claimed.

Final status

Florence demonstrates a functioning design pattern for a multimodal Telegram assistant backed by a RAG knowledge base. The supplied transcript confirms useful successful responses, but also reveals unresolved reliability issues. The appropriate project status is **prototype / iterative development**, not production-ready. The next best step is a repeatable regression test pass after addressing unsupported claims, context relevance and image-question completeness.

End of documentation. Prepared from user-supplied project artifacts and chat transcript; no claims of live deployment or external verification are made.